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26 SEPTEMBER 1960
SUPERSEDING
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MILITARY SPECIFICATION

CHARGER, BATTERY, METALLIC (POWER UNIT PP–34( )/MSM)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the requirements for Rectifier Power Unit PP–34( )/MSM which is designed to convert 105–125 volts, 50–60 cycle, single phase (a. c.) power to cover from 2 to 30 volts (d. c.) range and capable of charging from 1 to 12 lead-acid type storage cells at a maximum charging rate of 50 amperes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

T–T–871 —Twine, Cotton, Wrapping.
QQ–S–571 —Solder: Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.
UU–T–111 —Tape; Paper, Gummed (Sealing and Securing)
PPP–B–585 —Boxes; Wood, Wirebound.
PPP–B–591 —Boxes; Fiberboard, Wood-Cleated.
PPP–B–601 —Boxes; Wood, Cleated-Plywood.
PPP–B–621 —Boxes; Wood, Nailed and Lock-Corner.
PPP–B–636 —Boxes; Fiber.
PPP–P–291 —Paperboard, Wrapping, Cushioning.
PPP–T–76 —Tape, Pressure-Sensitive, Adhesive, Paper, Water-Resistant.

MILITARY

MIL–P–116 —Preservation, Methods of.
MIL–B–117 —Bags, Interior Packaging.
MIL–B–131 —Barrier Material; Water Vaporproof, Flexible.
MIL–T–152 —Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL–D–3464 —Desiccants, Activated (in Bags) for Static Dehumidification and Packaging.
MIL–B–10377 —Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL–F–14072 —Finishes for Ground Signal Equipment.
MIL–D–70327 —Drawings, Engineering and Associated Lists.

STANDARDS

MILITARY

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.
MIL–STD–130—Identification Marking for U. S. Military Property.

FSC 6130

571045–60–(536)

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MIL–STD–170—Moisture Resistance Test Cycle for Ground Signal Equipment.

MS20003 —Indicator, Humidity, Card, Three-Spot Impregnated Areas (Cobaltous Chloride).

DRAWINGS

SIGNAL CORPS

SC–DL–19439 —Rectifier Power Unit PP–34( )/MSM Drawing and Data List.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classificatiton

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

CONSOLIDATION CLASSIFICATION COMMITTEE

Uniform Freight Classification

Consolidated Freight Classification

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Description. Rectifier Power Unit PP–34( )/MSM converts 105 to 125 volts, single phase, alternating current (a. c.) into direct current (d. c.) for charging lead-acid type storage batteries.

3.2 Construction. The equipment shall be constructed in accordance with Drawing and Data List SC–DL–19439.

3.3 Preproduction samples. The contractor shall furnish preproduction samples for approval, as required by the invitatiton for bids and contract. (See 6.2(e)(1)).

3.4 Soldering.

3.4.1 *Solder.* Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ–S–571.

3.4.1.1 *Flux and cleaning agents.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.4.1.2 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Wherever practical, excess rosin shall be removed by the use of wire brush and then a dry cloth; any resulting loose flakes of rosin shall be carefully removed from the inside of the equipment. Insulating material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4.1.3 *Cleaning.* Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.5 Welding. Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded part. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. Wherever possible, the welded joint shall be such that grinding on the finished weld will be unnecessary.

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3.5.1 *Cleaning.* Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.5.2 *Process.* Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable for welding of aluminum or magnesium.

3.6 Tropicalization.

3.6.1 *Treatment of assembled equipment.* Assembled equipment shall be treated in accordance with Specification MIL–T–152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc.) which is applicable on contract.

(2) It does not incorporate any material listed in 3.6.3 unless the material has been given the treatment specified herein or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL–F–14072.

3.6.2 *Treatment of materials.*

3.6.2.1 *Treating materials.* Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.6.2.2 *Toxicity.* Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

3.6.2.3 *Flexibility.* Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.6.2.4 *Statement of treatment.* The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment is treated in accordance with Specification MIL–T–152, the above statement shall be incorporated in the statement of treatment required by that specification.

3.6.3 *Materials not resistant to moisture and fungi.* The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.
(b) Cotton.
(c) Cork.
(d) Felt, hair or wool.
(e) Jute.
(f) Leather.
(g) Linen.
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen or wood-flour as a filler or base.

3.7 Electrical and mechanical requirements.

3.7.1 *Input.* Charger PP–34( )/MSM shall be tested in accordance with 4.6.3 and shall be capable of operation from an input of 105 to 125 volts, 50–60 cycles, single phase alternating current.

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3.7.2 *Output.*

3.7.2.1 *Voltage.* Charger PP–34( )/MSM shall be tested in accordance with 4.6.3 and shall be capable of supplying a variable output voltage from 2 to 30 volts d.c.

3.7.2.2 *Current.* Charger PP–34( )/MSM shall be tested in accordance with 4.6.2 and shall supply a variable charging current from 1 to 50 amperes in approximately ½ ampere steps with fine control switch and 5 ampere steps with coarse control switch.

3.7.3 *Efficiency.* The efficiency of Charger PP–34( )/MSM shall be tested in accordance with 4.6.4 and shall not be less than 50 percent when delivering 50 amperes through a resistive load.

3.8 Operation. The Charger PP–34( )/MSM shall be tested in accordance with 4.6.3 and shall be capable of charging from 1 to 12 lead-acid storage cells connected in series.

3.9 Service conditions. The equipment shall meet the following service conditions:

3.9.1 *Equipment nonoperating.* The equipment shall meet the requirements of 3.7 after subjection to any one or more of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit:

(a) Temperature. Exposure in the range of +160° F. to —80° F.; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours at any one time at any orientation. (See 4.6.7).

(b) Relative humidity. When tested in accordance with 4.6.8 relative humidity up to 97 percent, for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours. Following subjection to such conditions and a power-on period of 16 hours at room temperature and relative humidity, the equipment shall meet the requirements of 3.9.2.

(c) Elevation. Elevation up to 25,000 feet above sea level.

(d) Orientation. Indefinite storage in any orientation.

(e) Vibration, bounce. As described in 3.10.

3.9.2 *Equipment operating.* The equipment shall meet the performance requirements of this specification when subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operation:

3.9.2.1 *Cold temperature.* Charger PP–34( )/MSM shall be tested in accordance with 4.6.5 and shall be capable of delivering a maximum charging current of 50 amperes at an ambient temperature of —40° F.

3.9.2.2 *Dry heat.* Charger PP–34( )/MSM shall be tested in accordance with 4.6.6 and shall be capable of delivering a maximum charging current of 40 amperes at an ambient temperature of +130° F.

3.9.2.3 *Relative humidity.* The equipment shall be tested in accordance with 4.6.8 and shall operate satisfactorily under all conditions of relative humidity.

3.9.2.4 *Altitude.* Charger PP–34( )/MSM shall be tested in accordance with 4.6.9 and shall meet the requirements of 3.7.2.2.

3.10 Vibration and bounce.

3.10.1 *Vibration.* Charger PP–34( )/MSM shall be tested in accordance with 4.6.10 and shall meet the requirements of 3.7 and there shall be no damage to the equipment.

3.10.2 *Bounce.* Charger PP–34( )/MSM shall be tested in accordance with 4.6.11 and shall meet the requirements of 3.7 and incur no physical damage except surface abrasion. There shall be no degradation in performance.

3.11 Preconditioning. Equipment shall be capable of meeting the inspection of section 4 without subsequent processing after subjection to the bounce preconditioning of 4.5. (Also see 4.4 (d)).

3.12 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished as

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specified by the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order. (See 6.2.)

3.13 Identification of product. Equipment assemblies, and parts shall be marked for identification in accordance with Standard MIL–STD–130 in addition to any other identification marking specified by the contracting officer.

3.14 Interchangeability. Corresponding units and replaceable assemblies, subassemblies, and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL–D–70327 shall govern changes in the manufacturer's part number. (See 4.9.) When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular assembly, subassembly, or part is to be considered replaceable, the contracting officer shall be consulted.

3.15 Workmanship. Workmanship for this equipment shall be such as to meet all applicable requirements of this specification, when inspected in accordance with 4.6.1 including the following:

3.2 Construction.
3.4 Soldering.
3.5 Welding.
3.6 Tropicalization.
3.13 Identification of product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility. Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3).

(b) Acceptance inspection. (Acceptance inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.4 and 4.10.)

(1) Acceptance inspection of items before preparation for delivery. (See 4.4).

(2) Acceptance inspection of preparation for delivery. (See 4.10 and section 5).

4.2.1 *Inspection conditions and methods.* Unless otherwise specified herein, all inspection shall be made at room ambient temperature, pressure and humidity.

4.2.2 *Inspection equipment and facilities.* Inspection equipment and facilities shall be of sufficient accuracy, quality, and quantity to permit the performance of the required inspection. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

4.3 Preproduction inspection. This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I and the group A, group B, group C inspection specified in tables II, III, and IV respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

TABLE I. *Preproduction Inspection*

Inspection	Requirement paragraph	Inspection paragraph
Vibration	3.10.1	4.6.10
Humidity	3.9.2.3	4.6.8
Bounce	3.10.2	4.6.11
Altitude	3.9.2.4	4.6.9

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TABLE II. *Group A Inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and mechanical	3.15	4.6.1	1.5 dphu	6.5 dphu
Output current test	3.7.2.2	4.6.2	1.5 dphu	*

* All defects for output test considered major.

4.4 Acceptance inspection of items before preparation for delivery. The contractor shall perform the inspection specified by (a) through (d) below, to demonstrate compliance with specified requirements. However, this does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government—at its discretion—may perform all or any part of the inspection specified in (a) to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection. (Also see 6.8).

(a) Acceptance inspection shall consist of group A, group B, and group C inspection as specified in 4.4.1 through 4.4.3.2.

(b) When Standard MIL–STD–105 specifies action by the Government, the Government may authorize the contractor to perform any of such actions except the responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when it is considered more practical to select separate samples for group B or group C inspection, or both.

(d) Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly. (See 3.11).

4.4.1 *Group A inspection.* Group A inspection (including sampling) shall conform to table II and Standard MIL–STD–105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

4.4.2 *Group B inspection.* Group B inspection (including sampling) shall conform to table III and the Appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.4.4 and the requirements of Standard MIL–STD–105 for disposition of rejected product.

4.4.2.1 *Group B sampling plans.* The group B sampling plans for the AQL's listed in table III shall be as follows:

AQL	*Inspection level for normal and tightened inspection*	*Inspection level for reduced inspection*
6.5%	L–7	L–5

4.4.2.2 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

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TABLE III. *Group B Inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL's
Operation	3.8	4.6.3	6.5%
Efficiency	3.7.3	4.6.4	6.5%
Interchangeability	3.14	4.9	6.5%

4.4.3 *Group C inspection.* This inspection shall be listed in table IV.

4.4.3.1 *Sampling for inspection of equipment.* One battery charger for each group C inspection from first units produced, and one from each 100, or fraction thereof, produced thereafter.

4.4.3.2 *Noncompliance.* If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.5.)

TABLE IV. *Group C Inspection*

Inspection	Requirement paragraph	Inspection paragraph
Cold	3.9.2.1	4.6.5
Dry heat	3.9.2.2	4.6.6

4.4.4 *Disposition of nonconforming product.* The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault.

Sample units found defective during group A, B, or C inspection.

Inspection lots which failed during group A or B inspection.

Product which has been reworked as a result of failure during group C inspection.

The required information shall be made available to the Government when the product is submitted and shall become the property of the Government.

4.5 Bounce Preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.6 Methods of tests.

4.6.1 *Visual and mechanical inspection.* Charger PP–34( )/MSM shall be examined to verify that the materials, design, construction, physical dimension and workmanship are in accordance with the applicable requirements. (See 3.2, 3.4, 3.5, 3.6, 3.13.) Visual and mechanical defects shall be classified by the Government as major or minor in accordance with definitions of Standard MIL–STD–105.

4.6.2 *Output current test.* A resistive load of .48 ohms shall be connected across the output terminals of the charger and an input voltage of 115 volts, 60 cycles, single phase a.c. maintained during the test. When the coarse and fine control switches are turned to each successive position the charger shall meet the requirements of 3.7.2.2.

4.6.3 *Charging test.* Charger PP–34( )/MSM shall be operated with input voltages of 105, 125, and 115 volts, 60 cycles, single phase a.c. applied as specified in steps (a) to (h) below. Connect two fully discharged lead-acid 8T type, 12 volt, 200 ampere batteries (specific gravity of electrolyte 1.140 ± .010), in series across the output terminals and charge as follows:

	Input	*No. cells*	*Charging rate*	*Voltage*	*Time*
a.	105 volts	1	1 ampere	--	5 minutes
b.	125 volts	1	1 ampere	--	5 minutes
c.	105 volts	12	50 ampere	--	5 minutes
d.	125 volts	12	50 ampere	--	5 minutes
e.	115 volts	12	50 ampere	--	5 minutes
f.	115 volts	12	10 ampere	--	19 hrs, 35 minutes or until batteries reach full charge
	(Batteries are fully charged when the electrolyte measures 1.270±.010 specific gravity)				
g.	115 volts	1	1 ampere	2	1 minute
h.	115 volts	1	50 ampere	30	1 minute

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The charger shall meet the requirements of 3.8.

4.6.4 *Efficiency test.* A resistive load of .48 ohms shall be connected across the output terminals of the charger and an input voltage of 115 volts, 60 cycles, single phase a.c. applied. The charger shall meet the requirements of 3.7.3. (Use the following formula):

$$\frac{\text{output (watts)}}{\text{input (watts)}} \times 100\% = \text{Efficiency}$$

4.6.5 *Cold test.* Connect the charger to an input voltage of 115 volts, 60 cycles, single phase a.c. and connect a resistive load of .48 ohms across the output terminals of the charger. The charger only, (not operating), shall be placed in the cold chamber with the resistive load outside and the chamber temperature maintained at —40° F. for a period 24 hours. At the end of this period the charger shall be operated for 4 hours and shall meet the requirements of 3.9.2.1.

4.6.6 *Dry heat test.* Connect the charger to an input voltage of 115 volts, 60 cycles, single phase a.c. and connect a resistive load of .60 ohm to the output terminals of the charger. The charger only, (not operating), shall be placed in the dry chamber with the resistive load outside and the chamber temperature maintained at +130° F. for a period of 24 hours. At the end of this period the charger shall be operated for 4 hours and shall meet the requirements of 3.9.2.2. The thermal protective unit (TH.1) should be shorted during this test only.

4.6.7 *Storage test.* The charger PP–34( )/MSM shall be given the storage test as required in 3.9.1(a). At the end of this test the charger shall meet the requirements of 3.7.2.2.

4.6.8 *Moisture-resistance test for equipment.*

4.6.8.1 *Test condition.* The charger PP–34( )/MSM shall placed in a test chamber.

(a) The charger shall not be removed from the test chamber for measurements.

(b) Start measurements not more than 5 minutes after power has been applied to the samples. Complete measurements as rapidly as possible. Do not leave power on the equipment after measurements have been completed.

4.6.8.2 *Test procedure.* The sample shall be tested as follows:

(a) Dry sample at 130 ± 5° F. for 24 hours.

(b) Measure the performance of the samples and realine or readjust if necessary, to meet requirements of this specification.

(c) Subject the samples to continuous cycling for a total of 5 of the 48-hour cycles shown on Standard MIL–STD–170 and measure their performance during the periods indicated on the standard.

4.6.9 *Altitude test.* The charger connected to an input voltage of 115 volts, 60 cycles, single phase a.c. and adjusted to deliver 50 amperes through a resistive load of .48 ohms connected to the output terminals shall be placed in an altitude chamber in operating condition. The barometric pressure shall be lowered to 20.6 inches of mercury and maintained at this level for ½ hour. Remove charger from chamber and test in accordance with 4.6.2. (See 3.7.2.2.)

4.6.10 *Vibration.* The vibration test shall be as follows:

(a) Rectifier Power Unit PP–34( )/MSM shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. The vibration table shall provide approximately sinusoidal vibration.

(b) The equipment shall be vibrated successively for one hour, in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per-second steps. The total excursion shall be $\frac{1}{64}$ inch. The test shall not exceed 3 hours.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be de-

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termined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because the unit weighs more than 250 pounds or because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.6.11 *Bounce test; equipment under 1000 pounds.* The equipment shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1 percent, for a total of 3 hours. After each ¾ hour period of the test, the equipment shall be repositioned by rotating it 90° clockwise in the same phase. (See 3.10.2.)

4.7 Inspection for corrosion. This inspection, covered by 4.6.1 shall be performed before and after 4.6.7 and 4.6.8. Upon inspection any evidence of corrosion shall be considered as a failure. (See FINISH PROTECTIVE, Specification MIL–F–14072 as per Drawing and Data List SC–DL–19439.)

4.8 Failure. If any equipment fails to meet the tests of 4.6.2 during cycling it shall be readjusted once. If the equipment then fails to meet the test of 4.6.2 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after condition and adjustment in 4.6.7 and 4.6.8 it does not pass the test.

4.9 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.14. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.10 Acceptance inspection of preparation for delivery. The contractor shall perform the inspection specified in 4.10.1 and 4.10.2 to demonstrate compliance with specified requirements. However, this does not relieve the contractor of responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government—at its discretion—may perform all or any part of the inspection specified in 4.4.1 and 4.4.2, to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection.) (Also see 6.5). When Standard MIL–STD–105 specifies action by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.10.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as spec-

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TABLE V. *Classification of Defects for Preservation, Packaging and Marking*

Major	Minor
1. Use of improper or defective material. 2. Quantity in unit package not as specified. 3. Incorrect packaging method applied. 4. Cushioning or padding omitted. 5. Cushioning or padding inadequate for the protection of the barrier material from projections, sharp edges, or other similar features of the item. 6. Cushioning inadequate for the physical and mechanical protection of the item. 7. Desiccant omitted. 8. Desiccant not in proper quantity to meet minimum requirements. 9. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope. 10. Stock number omitted, incorrect, or illegible. 11. Nomenclature omitted, incorrect, or illegible. 12. Marking of quantity of items in package omitted, incorrect, or illegible.	1. Item not properly blocked or braced within the unit package to prevent movement. 2. Humidity indicator omitted. 3. Packaging material damaged. 4. Corners of inner carton not blunted. 5. Air not expelled from barrier. 6. Conforming wraps are not snug-fitting and contain voids. 7. Desiccant not properly secured. 8. Insufficient area of barrier material provided to permit 2 subsequent seals. 9. Any item of marking information other than 10 thru 12 omitted, incorrect, or illegible.

ified in Specification MIL–P–116. Classification of defects shall be as shown on table V.

4.10.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL–STD–105. Inspection level L–8 shall be used for normal inspection and L–6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table VI.

4.10.2.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.10.2.2 *Disposition of nonconforming product.* Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL–STD–105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.11 Rough handling test (preparation for delivery). When rough handling test, in accordance with Specification MIL–P–116, is required by the contract (see 6.2(c)), the test described in 4.6.3 shall be conducted to determine freedom from operational malfunction caused by the rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Charger Battery, Metallic (Power Unit PP–34 ( )/MSM) shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and as described herein.

5.1.1.1 *Technical manual.* Each technical manual shall be packaged Method IC–3 as follows: Inclose each technical manual within a close-fitting type II class b bag conforming to

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TABLE VI. *Classification of Defects for Packing and Marking*

Major	Minor
1. Use of improper or defective material. 2. Quantity in pack not as specified. 3. Box closure not as specified. 4. Type, grade, class, and style of the shipping container not as specified. 5. Strapping omitted. 6. Strapping inadequate or incorrectly applied. 7. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage. 8. Shipping documents or packing list omitted. 9. Stock number omitted, incorrect, or illegible. 10. Nomenclature omitted, incorrect, or illegible. 11. Marking of quantity of items in pack omitted, incorrect, or illegible. 12. Destination marking omitted, incorrect, or illegible. 13. Service designation (color marking) omitted. 14. Special marking or labeling (when required) omitted. 15. Overseas code marking omitted, incorrect, or illegible.	1. Unsealed carton. 2. Defective taping or sealing of carton. 3. Any other box defect which may be considered minor by definition of Standard MIL–STD–105. 4. Any item of required marking information other than 9 thru 15 listed under major defects omitted, incorrect, or illegible.

Specification MIL–B–117. Heat seal the bag to effect closure.

5.1.1.2 *Leads.* The lead shall be packaged Method III as follows: Group the leads together within type I class 1 paperboard cushioning material conforming to Specification PPP–P–291, making a snug conforming wrap. Seal the wrap with class 1 gummed paper tape conforming to Specification UU–T–111.

5.1.1.3 *Cable assembly.* Each cable assembly shall be packaged Method III as follows: Wind each cable assembly into a coil of smallest practicable dimensions, and tie at three places with type II cotton twine, conforming to Specification T–T–871, securing the free ends of the cable assembly to the coil with the ties.

5.1.1.4 *Consolidated package.* Consolidate the items packaged as specified in 5.1.1.2 and 5.1.1.3 together within a close-fitting one piece folder, fabricated of double faced corrugated fiberboard meeting the applicable provisions for class 1, Specification PPP–B–636. Closure shall be in accordance with the referenced specification.

5.1.1.5 *Unit packaging.*

5.1.1.5.1 *Charger, Battery, Metallic (Power Unit PP–34 ( )/MSM).* Each charger shall be unit packaged Method IIb as follows: Cushion each charger on all surfaces with cells or pads or both, fabricated of double faced corrugated fiberboard, and designed to absorb the shock of impact encountered in handling and transit. Fiberboard for cells and pads shall have dry bursting strength equal to the fiber box in which the cushioned item will be placed. Place the cushioned charger and consolidated package (see 5.1.1.4) together with the required number of units of activated desiccant within a close-fitting, regular slotted style corrugated fiber box, class 1, conforming to Specification PPP–B–636. The desiccant shall conform to Specification MIL–D–3464, and in the quantity in units as required by the formula in Specification MIL–P–116. The desiccant shall be positioned and secured uniformly through the package. The corners of the fiber box shall be blunted, and any stapled or stitched outer joints shall be covered with class 3 gummed paper conforming to Specification UU–T–111, except where the manufacturer's or body joint is the butted type. Box closure shall be by means of tape in accordance with Specification PPP–B–636. The sealed box shall be placed within a bag fabricated of water-vaporproof barrier material conforming to Specification MIL–B–131, class 1. One dimension of the bag shall be fabricated to permit 2 reseal-

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ings for inspection purposes. A humidity indicator card conforming to MS20003 shall be placed and secured on top of the interior box within the barrier beneath the sealed seam. Heat seal the barrier bag leaving a small opening to exhaust the air. Exhaust the air to a practicable minimum and complete the heat seal. The time lapse between positioning the desiccant and completion of sealing operation should be held to an absolute minimum. Place the water-vaporproofed charger within a close-fitting fiber box, V3c class 2, conforming to Specification PPP–B–636. Stapling or stitching on the inside of the box shall be covered with water-resistant pressure sensitive tape conforming to Specification PPP–T–76, except when the manufacfacturer's or body joint is butted type. Box closure shall be in accordance with the appendix of the referenced box specification.

5.1.2 *Level C.* Charger Battery, Metallic (Power Unit PP–34( )/MSM) shall be preserved and packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* Charger, Battery, Metallic (Power Unit PP–34( )/MSM) packaged as specified in 5.1.1 shall be packed for shipment one each within a close-fitting class 2 style 2 nailed and lock-corner wood box conforming to Specification PPP–B–621, or a class 3 style 2 wood wirebound box, conforming to Specification PPP–B–585, or a style A, type I class 1 wood, cleated plywood box, conforming to Specification PPP–B–601, or a style A wood-cleated fiberboard box conforming to Specification PPP–B–591, or an overseas type style A paper overlaid, veneer wood cleated box conforming to Specification MIL–B–10377. Box closure and strapping shall be in accordance with the applicable box specification or appendix thereto.

5.2.2 *Level B.* Charger, Battery, Metallic (Power Unit PP–34( )/MSM) shall be packed for shipment as specified in 5.2.1 except boxes shall be domestic type or class. No strapping required.

5.2.3 *Level C.* Charger, Battery, Metallic (Power Unit PP–34( )/MSM) shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classificatiton No. 4 and Consolidated Freight Classification Rule 21 for rail shipment, National Motor Freight Classification No. A–3 and No. 14 for truck shipment, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL–STD–129.

6. NOTES

6.1 Intended use. Rectifier Power Unit PP–34( )/MSM is used to convert 105–125 volts, 50 to 60 cycle, single-phase a.c. power to direct current suitable for charging 1 to 12 lead-acid type storage batteries.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(c) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample.

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon—including rough handling test, which will not be performed as procurement inspection.

d. Five copies of process sheet, using format shown on Drawing SC–C–33073 and accompanied by such

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drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to indentify the items herein. These data should be submitted to the contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required. (See 3.12.)

(g) Quantity of tools and running spare parts required.

(h) Submission of the statement of treatment referenced in 3.6, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Contractual requirements. It is recommended that the following be included in the contract:

6.3.1 *Additional testing.* It should be understood that additional inspection and nondestructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable documents. The Government inspector, therefore, may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection and testing, performed either at a Government laboratory or the contractor's plant.

6.3.2 *Waiver of proproduction samples.* When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that the specified samples have been submitted and approved on the current contract; production is in strict compliance with specified requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example, PP–34W/MSM. As soon as possible after the award of contract, the contractor should apply to the contracting officer for such information. (See 1.1.)

6.5 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.4.3.2.)

6.6 Definition of level B. Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.7 Inspection. Inspection is the examination or testing, or both, of samples to determine compliance with applicable requirements. Sampling is an element of inspection.

6.7.1 *Examination.* Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.7.2 Testing consists of determinations of compliance using technical means.

6.8 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definite Quality Assurance Provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A. The total of that inspection set forth in the Quality Assurance

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Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Standard MIL–STD–105.

(b) Type B. That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Standard MIL–STD–105.

(c) Type C. A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Signal Corps
Navy—Marine Corps
Air Force

Preparing Activity:

Army—Signal Corps

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